

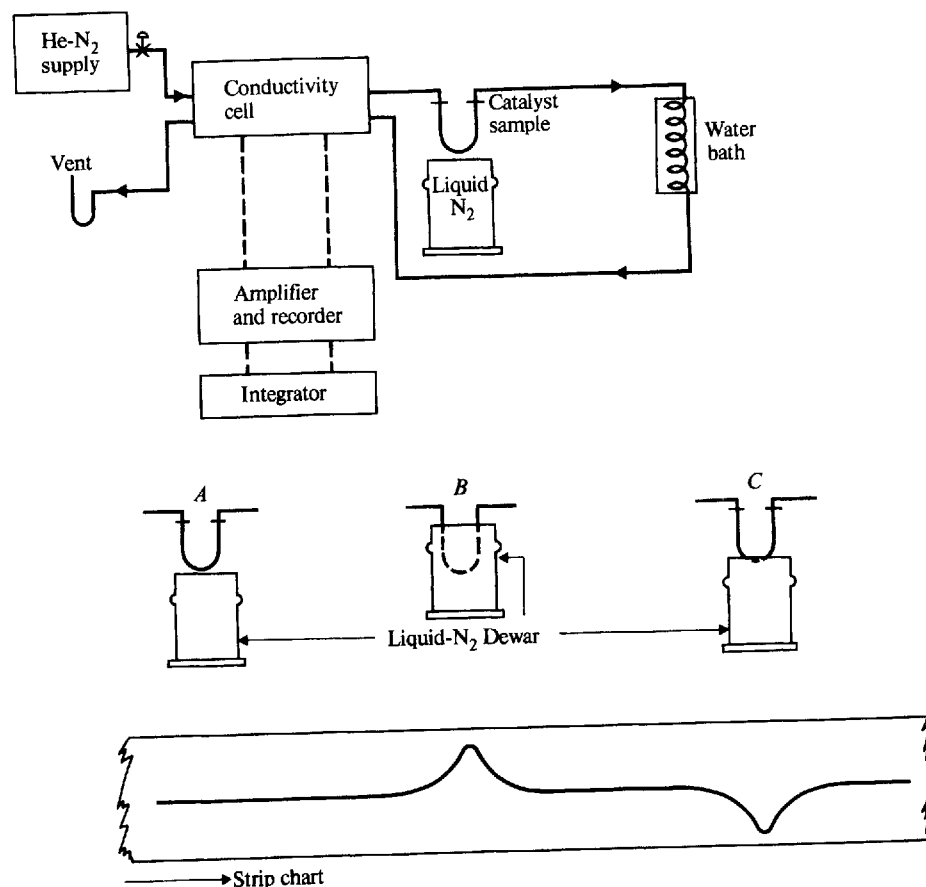


FIGURE 8-3
Flow BET apparatus and technique, also applicable to chemisorption at any temperature.

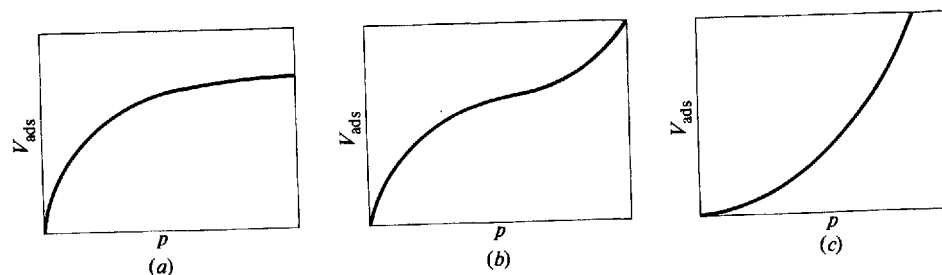


FIGURE 8-4
Three types of physisorption isotherms: (a) Langmuir, type I, (b) type II (c) type III.

Types I and II are rather common, while others arise under somewhat exceptional circumstances.

8-9 THE MULTILAYER-ADSORPTION THEORY (BET EQUATION)

In 1938 Brunauer, Emmett, and Teller¹ published the paper Adsorption of Gases in Multimolecular Layers, in which a generalized theory of physical adsorption was set forth in some detail. A signal feature of this development was a clear specification of the volume of gas adsorbed at the point of monolayer completion. In consequence, the means became available for computing with rigor the total surface area of a porous body, e.g., catalyst, from physical-adsorption equilibrium data. Adsorption equilibrium models proposed before the BET theory failed to yield an unequivocal means for determining monolayer coverage. As Sir Hugh Taylor put it, "The technique [BET] is a standardized one which can be reproduced by different workers in different laboratories with considerable accuracy and reproducibility."

It is perhaps safe to say that the BET development was sponsored more by concern for catalyst-surface-area determination than for an adsorption-process rationale. Obviously a model of physical adsorption was required to facilitate a meaningful relationship between surface area and measured quantities. Earlier Emmett and Brunauer had described the use of adsorption isotherms for the determination of surface areas of various iron catalysts in ammonia synthesis. For Benton² had pointed out that in systems exhibiting type II isotherms the point of monolayer completion might correspond to that region of the isotherm where linearity starts. It was reasoned that the linear portion represented second-layer development; in consequence this point of initial multilayer formation would correspond to monolayer completion. This point was termed *point B* by Emmett and Brunauer. Other criteria had been suggested; however, point *B* proved to be the most reliable of proposed candidates.

In systems not characterized by a type II isotherm, the point-*B* method obviously fails. The physical adsorption of *n*-butane on silica gel is such an example. From 0 to about 300 mm linearity is manifest, thus making it impossible to specify point *B*. It was clear to Emmett and Brunauer that a generalized theory for physical adsorption was required which would prove more powerful, less ambiguous, and of broader scope in the determination of surface area.

The BET equation can be developed, first, for the case of adsorption upon a free surface so that no limit upon the number of layers of adsorbed gas is imposed. The integration can also be carried out for *n* layers, this corresponding to bounded surfaces (cracks, crevices, pores). In principle the derivation amounts to an extension of Langmuir's theory (monolayer adsorption) to multilayer adsorption.

¹ S. Brunauer, P. H. Emmett, and E. Teller, *J. Am. Chem. Soc.*, **60**: 1309 (1938).

² See Emmett, loc. cit.

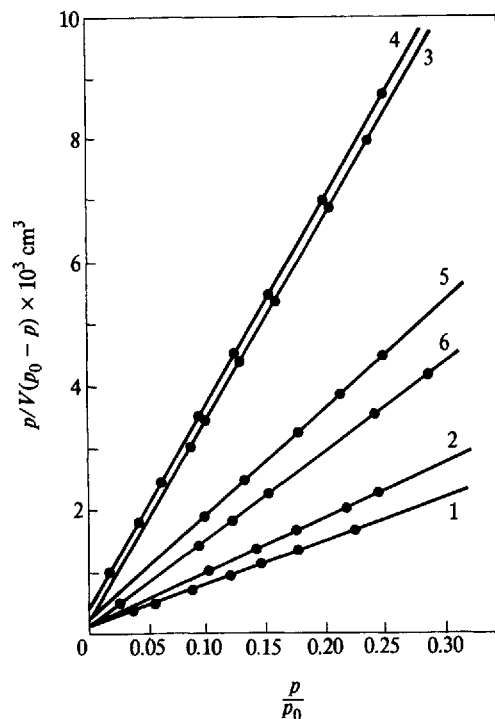


FIGURE 8-5
Linearization of physisorption data according to the BET equation (8-17). Adsorption of N_2 at 90.1 K on unpromoted Fe catalyst 973 per 489.0 g (curve 1); Al_2O_3 -promoted Fe catalyst 424 per 49.8 g (curve 2); Al_2O_3 - K_2O -promoted Fe catalyst 958 per 54.4 g (curve 3); fused-copper catalyst per 550.0 g (curve 4); chromium oxide gel per 1.09 g (curve 5); and silica gel per 0.606 g (curve 6). [S. Brunauer, P. H. Emmett, and E. Teller, *J. Am. Chem. Soc.*, **60**: 309 (1938).]

which is the Langmuir isotherm, while as p approaches p_0 , the volume adsorbed versus p/p_0 displays the convex character typical of type II isotherms at high pressures.

Equation (8-15) is readily cast into a form convenient for the abstraction of V_m

$$\frac{p}{V(p_0 - p)} = \frac{1}{V_m c} + \frac{c - 1}{V_m c} \frac{p}{p_0} \quad (8-17)$$

A plot of the left-hand side (experimental data) against p/p_0 yields a straight line from which both V_m and c can be obtained by employing the slope and intercept (Fig. 8-5).

The free-surface model is an idealization, as most solid catalysts contain cracks, crevices, and pores, within which the major fraction of the surface area resides. Only n layers of gas can be adsorbed within a pore or crevice, so that the upper limit of the summations in Eq. (8-13) should be specified as n rather than infinity. In this instance

$$\frac{V}{V_m} = \frac{xc}{1-x} \frac{1 - (n+1)x^n + nx^{n+1}}{1 + (c-1)x + cx^{n+1}} \quad (8-18)$$

For $n = 1$

$$\frac{V}{V_m} = \frac{xc}{1-x} \frac{1 - 2x + x^2}{1 + (c-1)x - cx^2} = \frac{xc}{1-x} \frac{(1-x)^2}{(1+cx)(1-x)} \quad (8-19)$$

or

$$\frac{V}{V_m} = \theta = \frac{cx}{1+cx}$$

which is the Langmuir isotherm. Since x is less than unity, it is apparent that as n approaches infinity, Eq. (8-18) reduces, as expected, to Eq. (8-15).

Further Remarks on the BET Equation

Given data on volume adsorbed vs. pressure, the BET equation (8-17) provides a reliable means whereby total surface area can be determined. The method of data procurement is irrelevant with respect to application of Eq. (8-17), though it is certainly significant that more data are obtained in a given period by the Shell flow method, cited earlier. Whether static or flow techniques are employed, inspection of the BET relation suggests that since the constant c is of the order of 10^2 for N_2 , one low-pressure point on the isotherm may be sufficient for a rather accurate surface-area determination. Thus a single adsorption equilibrium point in the range of p/p_0 of 0.2 to 0.3, when extrapolated to zero V at $p/p_0 = 0$, will yield a slope not significantly different from the true value obtained using several relative pressures. This follows from the fact that the intercept term, $1/V_m c$, will approach zero for most systems displaying a type II isotherm with a nitrogen adsorbate.

Pore Size and Its Distribution

In Fig. 8-2 a typical physical adsorption-desorption isotherm is shown for a porous solid. Hysteresis is evident, i.e., the desorption curve lies above that of adsorption equilibrium from p/p_0 of from unity to some intermediate value. In effect, then, adsorbate is retained during desorption and released at a p/p_0 value less than that required to cause adsorption. This behavior and its rationalization provides the basis for estimating pore size and its distribution. The phenomenon is due to capillary condensation-evaporation processes.

Lord Kelvin recognized that the vapor pressure of a liquid contained in a small-diameter capillary is less than the normal value predicted for a free surface. Consider a capillary partially filled with a liquid of surface tension σ . The change in free energy due to evaporation of a differential volume of liquid $n \Delta G$ equals the change in surface times the surface tension, where $n = dv/v_m$ and v_m is molar volume:

$$n \Delta G = + (2\pi r d\ell) \sigma \cos \theta = -nRT \left(\ln \frac{p}{p_0} \right)$$

$$n = \frac{dv}{v_m} = \frac{\pi r^2 d\ell}{v_m}$$

Then

$$\ln \frac{p}{p_0} = v_m \frac{2\pi r d\ell}{(\pi r^2 d\ell) RT} \cos \theta \approx 1$$

$$\frac{p}{p_0} = \exp \frac{-2\sigma v_m}{rRT} \quad (8-20)$$

which is the common form of the Kelvin equation, assuming a wetting angle of zero. This equation states that the pressure at which condensation or evaporation will occur is always less than the free-surface vapor pressure *if* the capillary radius is small enough for a given liquid. We see then that if a nonporous solid or one containing very large pores is subjected to physical adsorption-desorption equilibrium experiments, actual condensation of adsorbate will not occur until the gas pressure equals the vapor pressure at the prevailing temperature, i.e., at $p/p_0 = 1$ in Fig. 8-2. On the other hand, if pores of appropriate radius r exist, condensation will occur before $p/p_0 = 1$; thus liquid N_2 will form in the pores at $p/p_0 < 1$. On adsorption, this would account for the rapid increase in V with p/p_0 typical of adsorption isotherms for porous solids. However, this mechanism would not account for hysteresis if filling of pores on adsorption and their emptying on desorption followed the same mechanism. This physisorption mechanism is, of course, vertical filling and emptying of the pores. While various theories have been set forth to rationalize hysteresis, a very reasonable one due to Cohan¹ will be discussed here with the understanding that the matter is not yet totally resolved. Cohan argues that on *adsorption* the pores do not fill vertically but *radially*. So we see then that with condensation of the first layer, r is decreased, thus causing further condensation at a fixed p/p_0 . In other words pores of a radius r , corresponding to a given p/p_0 , fill instantaneously. The Kelvin equation can be derived for this situation, where now the change in volume dv is $2\pi r \ell dr$ and that of surface is $2\pi \ell dr$. Balancing the free-energy change as above, we find

$$\frac{p}{p_0} = \exp \frac{-\sigma v_m}{rRT} \quad (8-21)$$

Following Cohan, for a given pore radius r , adsorption with radial capillary condensation occurs at

$$p_{ads} = p_0 \exp \frac{-\sigma v_m}{rRT} \quad (8-22)$$

while *vertical* emptying of the pores occurs during desorption at about

$$p_{des} = p_0 \exp \frac{-2\sigma v_m}{rRT} \quad \text{implies wetting angle of zero} \quad (8-23)$$

It follows then that the adsorption pressure p_{ads} is related to that of desorption p_{des} by

$$p_{ads}^2 = p_0 p_{des} \quad (8-22a)$$

or the reduced pressure required to empty the capillary is equal to the square of that necessary to fill it. A pore of radius r which fills at p/p_0 of say 0.7 is desorbed at

¹ L. H. Cohan, *J. Am. Chem. Soc.*, **60**: 433 (1938), **66**: 98 (1944).

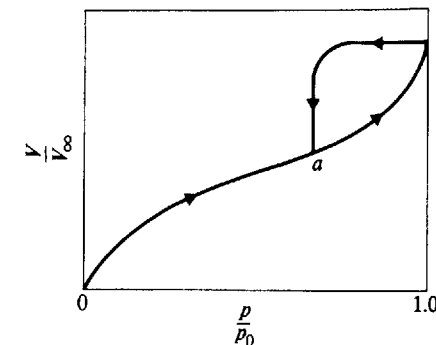


FIGURE 8-6

Isotherm for a porous solid of uniform pore radius. Point a is that of step change in V/V_∞ .

p/p_0 of about 0.5; hence hysteresis. Some data support this quantitative prediction.

Given the adsorption-desorption hysteresis loop, pore size and its distribution can, in principle, be obtained. If, for example, a porous catalyst sample contained pores of one diameter only, the hysteresis loop would resemble that shown in Fig. 8-6. When we know σ and v_m and that the desorption value of p/p_0 occurs at point a in Fig. 8-6, the pore radius can be calculated from the Kelvin equation. Rarely does the hysteresis loop prove to be as simple as that shown in Fig. 8-6; a less abrupt desorption curve is common, as in Fig. 8-2, suggesting a distribution of pore radii. The problem of determining the exact distribution is not unlike that encountered earlier, namely, finding an analytical residence-time distribution of flowing molecules which fits an experimental residence-time response curve for a vessel.

Wheeler¹ and Shull² set forth the general relationships which permit the evaluation of pore-size distribution from physical-desorption isotherms. The volume of gas not adsorbed at a pressure p is $V_s - V$, where V_s is total pore volume, i.e., volume of gas adsorbed at p_0 . This volume $V_s - V$ must equal the summation of volumes of unfilled pores of radii which may be distributed. Before a pore is filled, its effective radius is less than the true radius by the thickness t of the physically adsorbed multilayer. On emptying, of course, all liquid is evaporated except the physically adsorbed layer of thickness t . In the Kelvin equation the radius of interest is not r but $r - t$. Wheeler's reasoning suggests

$$V_s - V = \pi \int_{r_c}^{\infty} (r - t)^2 L(r) dr \quad (8-24)$$

i.e., the summation of all pore volumes composed of volumes $\sum \pi(r_i - t)^2 L(r) dr$, where $L(r) dr$ is the total length of pores of radii between r and $r + dr$. r_c is the critical pore radius (corrected for physical adsorption), which means that all pores of radii less than r_c are filled and those of greater radii are empty. The

¹ A. Wheeler, in P. H. Emmett (ed.), "Catalysis," vol. 2, Reinhold, New York, 1955.

² C. G. Shull, *J. Am. Chem. Soc.*, **70**: 1405 (1948).

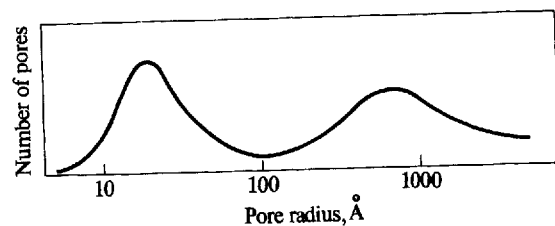


FIGURE 8-7
Bimodal pore-size distribution.

key to the use of Eq. (8-24) is the proper choice of $L(r) dr$. From the Kelvin equation

$$r_c = t + \frac{2\sigma v_m}{RT \ln(p_0/p)} \quad (8-25)$$

so that once $L(r) dr$ is specified, the Wheeler-Shull relation can be solved. In general for an assumed $L(r) dr$ function the V -versus- p/p_0 data and prediction are compared and $L(r) dr$ adjusted until conformity is realized between model and data. Barrett, Joyner, and Helenda¹ provide a detailed procedure for pore-size determination. In any case, t must be secured from experimental isotherms for non-porous surfaces. Anderson² provides an improvement of the method. Typical pore-size distribution is presented in Fig. 8-7 for a microporous-macroporous catalyst (Fig. 8-1). Note the bimodal distribution.

A Simplified Model of Average Pore Size

In contrast to Fig. 8-1, if only macropores characterize the catalyst, an estimation of average pore radius can be made by assuming a cylindrical pore of length L and radius r . The internal surface-to-volume ratio of this idealized pore is

$$\frac{S}{V_p} = \frac{2rL}{r^2L} = \frac{2}{r} \quad (8-26)$$

Hence a value of pore volume V_p and BET area S permits a rough estimate of average pore radius r by

$$r = \frac{2V_p}{S} \quad (8-27)$$

Obviously if this simplified model is applied to a macroporous-microporous catalyst sample exhibiting a bimodal distribution, a totally meaningless value of r will result. Values of S , V_p , and r for several catalysts are given in Table 8-2.

Table 8-2 SURFACE AREA, PORE VOLUME, AND MEAN PORE RADII FOR TYPICAL SOLID CATALYSTS†

Catalyst	Surface area S_g , m ² /g	Pore volume V_p , cm ³ /g	Mean pore radius, Å
Activated carbons	500-1500	0.6-0.8	10-20
Silica gels	200-600	0.4	15-100
SiO ₂ -Al ₂ O ₃ cracking catalysts	200-500	0.2-0.7	33-150
Activated clays	150-225	0.4-0.52	100
Activated alumina	175	0.39	45
Celite (kieselguhr)	4.2	1.1	11,000
Synthetic ammonia catalysts, Fe	4-11	0.12	200-1000
Pumice	0.38		
Fused copper	0.23		

† From J. M. Smith, "Chemical Engineering Kinetics," 1st ed., McGraw-Hill, New York, 1956. Some values from A. Wheeler, *Adv. Catal.*, 3: 250-326 (1950).

8-10 RELEVANCE OF CHEMISORPTION TO CATALYSIS

Attention was just directed to a physical process (physisorption) which is germane to catalysis insofar as knowledge of the physical area and pore distribution of catalyst particles can be derived from physical-adsorption isotherms. The relevance of chemisorption to catalysis is perhaps best stated in a general historical context.

When we recognize that catalysis developed as a branch of chemical kinetics, it is hardly surprising that early catalytic studies were conducted as were homogeneous chemical-kinetic studies. The procedure followed to realize an understanding of, say, homogeneous gas-phase reactions is today clearly recognized. As noted by Boudart¹ and Halsey,² the steps are as follows: (1) the overall reaction must be dissected into its most elemental steps in a sequence which yields the net rate then (2) analyzed theoretically and if possible experimentally. By way of illustration, the first-order decomposition of N₂O₅, long considered a simple process, was eventually shown (Ogg³) to be the result of a series of elemental steps which yield the overall result.

Analyses of catalytic reactions must follow this general procedure. However, difficulties become legion in the light of early catalytic kinetic studies. For overall catalytic rates could not be simply expressed in terms of measured gas-phase concentrations. As surface concentrations of reactants clearly determined surface (catalytic) rates, an additional body of data was required to establish the relationship between surface and fluid concentrations.

¹ M. Boudart, *Chem. Eng. Prog.*, 58: 73 (1962).

² G. Halsey, *J. Phys. Chem.*, 67: 2038 (1963).

³ R. A. Ogg, *J. Chem. Phys.*, 15: 337, 613 (1947).

¹ E. P. Barrett, L. G. Joyner, and P. P. Helenda, *J. Am. Chem. Soc.*, 73: 373 (1941).

² R. B. Anderson, *J. Catal.*, 3: 50 (1964).

That body of data is the chemisorption isotherm for the system when surface reaction is rate-controlling, thereby establishing a relationship between surface concentration and the observable gas-phase concentration. When, however, adsorption is rate-controlling, the desirability of chemisorption *rate* data becomes apparent. In terms of catalytic rate phenomena, then, chemisorption equilibria and rate studies are required for a rational formulation of surface kinetics. These kinetic and thermodynamic data then permit a clear definition of the overall catalytic rate and thus invite, in principle, the dissection of overall rate into likely elementary steps. Given the fact the surface is an active participant in the sequence of elementary steps, it is not difficult to appreciate why our understanding of surface catalysis lags behind that of homogeneous reactions. For while notable progress has been made in the physics of the solid state, crucial property and energetic differences are recognized as existing between bulk and surface regimes in the solid state. As catalysis involves primarily solid surfaces, we can appreciate the profound complexities involved in characterizing the surface of a solid even when bulk properties are well known or measurable.

In addition to its kinetic significance, chemisorption, by its specificity with respect to particular surfaces as opposed to total surface (determined by non-specific BET physisorption) permits, in principle, the measurement of catalytic sites existing upon a support matrix which itself does not invite chemisorption. Specific examples of catalytic (as opposed to total-surface-area) determination by chemisorption and/or titration will be cited later in this chapter.

8-11 CHEMISORPTION EQUILIBRIA AND KINETICS

At chemisorption equilibrium, Langmuir asserted that ideal surface coverage is

$$\theta = \frac{Kp}{1 + Kp} \quad (8-28)$$

where

$$K = \frac{k_{\text{ads}}}{k_{\text{des}}} \quad (8-29)$$

The adsorption and desorption rate coefficients, k_{ads} and k_{des} , are assumed to obey an Arrhenius functionality. In essence, activation energies and adsorption heats are presumed to be constant, independent of θ . These assertions imply surface ideality; i.e., heat of adsorption q is independent of coverage θ , as are the activation energies E_{ads} and E_{des} .

In point of fact, few gas-solid catalyst systems conform to the Langmuir postulates. Measured values of q as a function of coverage θ , indicate a definite decline in q with θ , from which one is justified in concluding that either (1) catalytic surfaces are energetically nonideal (heterogeneous) and/or (2) interaction forces are manifested as a function of coverage. In fact Langmuir anticipated non-ideality and suggested that in such circumstances

$$\theta = \sum_{i=1}^{i=n} \frac{K_i p}{1 + K_i p} \quad (8-30)$$

That is, the real surface might be considered as a summation of small discrete ideal surfaces. This suggestion proved to be seminal.

Real Surfaces

Chemisorption equilibria are often described by the Freundlich isotherm

$$\theta = Kp^{1/n} \quad n > 1 \quad (8-31)$$

or the isotherm of Temkin

$$\theta = \frac{1}{f} \ln Kp \quad (8-32)$$

while in numerous instances the kinetics of chemisorption follow the Elovich equation

$$\bar{r}_{\text{ads}} = k_{\text{ads}} p \exp(-g\theta) \quad (8-33)$$

and desorption rates tend to conform to the relation

$$\bar{r}_{\text{des}} = k_{\text{des}} \exp h\theta \quad (8-34)$$

Thus observations of chemisorption equilibria and kinetics on real surfaces stand in radical contrast to the ideal-surface models. We note that upon equating (8-33) with (8-34), the equilibrium relation conforms to that of Temkin; this relation, however, fails to predict values of θ of zero and unity in the limits of p , and while the Freundlich isotherm predicts a zero value of θ at p equal to zero, values of θ exceed unity with increasing p .

Real-Surface Models

Brunauer, Love, and Keenan¹ have profitably analyzed chemisorption kinetics and equilibria upon real, heterogeneous surfaces. They assume adsorption and desorption activation energies (and hence equilibrium heats of adsorption) to be linear functions of coverage θ :

$$\begin{aligned} E_{\text{ads}} &= E_{\text{ads}}^0 + \gamma\theta \\ E_{\text{des}} &= E_{\text{des}}^0 - \beta\theta \\ q &= q_0 - \alpha\theta \quad \text{where } \alpha = \gamma + \beta \end{aligned} \quad (8-35)$$

The physical bases for their assumption can be (1) surface heterogeneity, (2) adsorbed-species interaction, or (3) a combination of them.

For the rate of adsorption, assuming a continuum of ideal patches, we have

$$\bar{r}_{\text{ads}} = \int_0^\theta k_{\text{ads}} p (1 - \theta) d\theta \quad (8-36)$$

where

$$k_{\text{ads}} = \mathcal{A} \exp \frac{-(E_{\text{ads}}^0 + \gamma\theta)}{RT}$$

¹ S. Brunauer, K. S. Love, and R. G. Keenan, *J. Am. Chem. Soc.*, **64**: 751 (1942).

Then upon integration of (8-36) for $1 - \theta \rightarrow 1$ we have

$$\bar{r}_{\text{ads}} = \mathcal{A} p \frac{RT}{\gamma} \exp \frac{-E_{\text{ads}}^0}{RT} \left(\exp \frac{-\gamma\theta}{RT} - \exp \frac{-\gamma}{RT} \right) \quad (8-37)$$

which properly reduces to zero at $\theta = 1$ and achieves a maximum at $\theta = 0$. For large values of γ , (8-37) becomes

$$\bar{r}_{\text{ads}} = k_{\text{ads}} p \exp(-g\theta) \quad \text{where } g = \frac{\gamma}{RT} \quad (8-38)$$

the Elovich relation. By similar reasoning the desorption rate

$$\bar{r}_{\text{des}} = \int_0^\theta k_{\text{des}} \theta d\theta \quad (8-39)$$

becomes upon integration, assuming a linear $E_{\text{des}} - \theta$ function (at $\theta \rightarrow 1$) and a large value of β ,

$$\bar{r}_{\text{des}} = k_{\text{des}} \exp h\theta \quad \text{where } h = \frac{\beta}{RT} \quad (8-40)$$

Chemisorption equilibrium for the real surface characterized by a linear q -versus- θ relation is

$$\theta = \int_0^1 \frac{p \mathcal{A} \exp(q_0/RT - \alpha\theta/RT) d\theta}{1 + p \mathcal{A} \exp[q_0/RT - (-\alpha\theta/RT)]} = \int_0^1 \frac{p a_0 \exp(-\alpha\theta/RT) d\theta}{1 + p a_0 \exp(-\alpha\theta/RT)} \quad (8-41)$$

which, upon integration, yields

$$\theta = \frac{RT}{\alpha} \ln \frac{1 + a_0 p}{1 + a_0 p \exp(-\alpha/RT)} \quad (8-42)$$

Two limiting cases are of interest.

CASE 1 The change in q with θ is large; that is, α is large and $a_0 p \exp(-\alpha/RT) \ll 1$. Then

$$\theta = \frac{RT}{\alpha} \ln(1 + a_0 p) \quad (8-43)$$

If α is quite small, q varies negligibly with θ and the ideal Langmuir isotherm applies.

CASE 2 If $a_0 p \gg 1$,

$$\theta = \frac{RT}{\alpha} \ln a_0 p \quad (8-44)$$

which is the celebrated Temkin isotherm.

A significant consequence of (8-42) results upon expanding the logarithm and excluding all but the first term

$$\ln x \approx 2 \frac{x-1}{x+1}$$

Thus

$$\theta = \frac{RT}{\alpha} \ln \frac{1 + a_0 p}{1 + a_0 p \exp(-\alpha/RT)} \approx \frac{RT}{\alpha} \frac{a_0 p [1 - \exp(-\alpha/RT)]}{2 + a_0 p [1 + \exp(-\alpha/RT)]}$$

when α is large, $\exp(-\alpha/RT) \ll 1$; then

$$\theta = \frac{RT}{\alpha} \frac{a_0 p}{1 + a_0 p/2} \equiv \text{Langmuir isotherm form} \quad (8-45)$$

It should be noted that the error attending the approximation of the logarithm is only 10 percent at a value of $x = 3$.

We see then that an adsorption isotherm rooted in real-surface physics readily reduces to an ideal (Langmuir) form. If we had assumed an exponential q -versus- θ functionality, then, as shown by Halsey and Taylor,¹ a Freundlich isotherm would result, that is, $\theta = K(p)^{1/n}$. In fact since²

$$\frac{ax}{1+ax} \approx kx^{1/n} \quad n > 1 \quad (8-46)$$

the equivalence of the Langmuir, Freundlich, and Brunauer-Love-Keenan isotherms is demonstrated. The situation is displayed in Fig. 8-8.

While noting that diverse chemisorption kinetic and equilibria functionalities are, in fact, well approximated by the Langmuir model rooted in ideality, it is to be observed that distinct advantages reside in the Langmuir formulation when

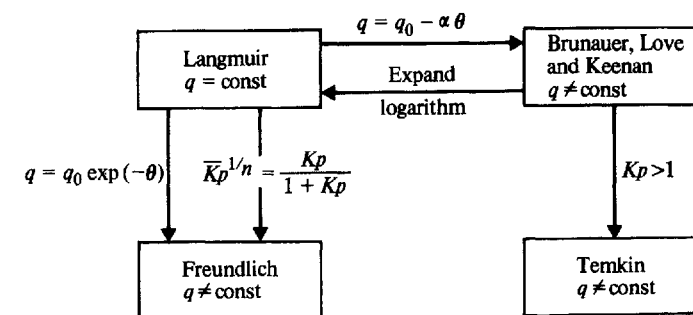


FIGURE 8-8
Equivalence of diverse adsorption equilibria.

¹ G. D. Halsey and H. S. Taylor, *J. Chem. Phys.*, 15: 624 (1947).

² M. Boudart, *AIChE J.*, 2: 62 (1956).